INCH POUND

MIL-C-3098/121C
24 September 1992
SUPERSEDING
MIL-C-3098/121B
8 March 1983

MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR132/U

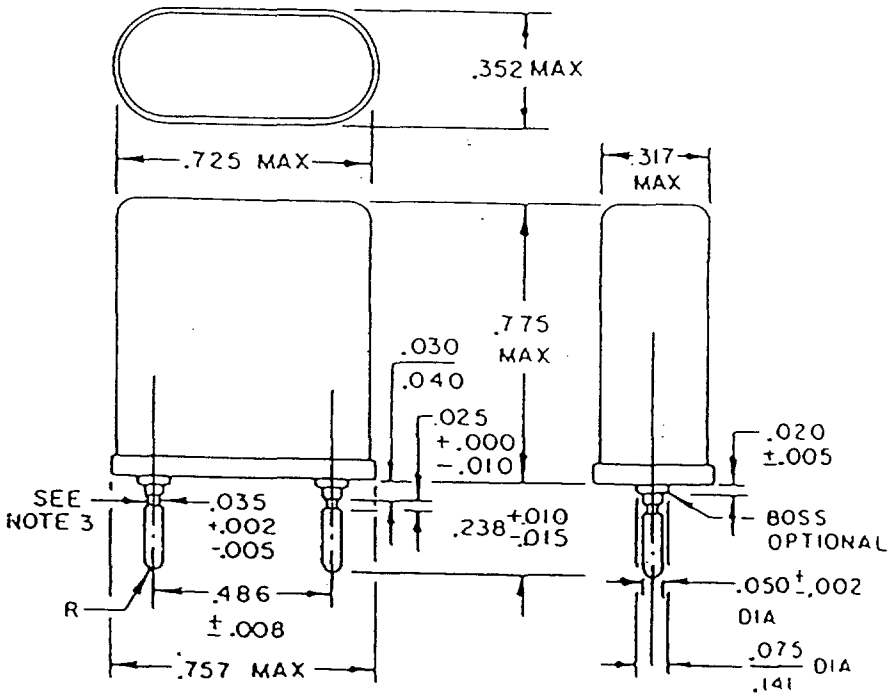
Inactive for new design

This specification is approved for use by all Department and Agencies of the Department of Defense.

The requirement for the acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 0.8 to 20 MHz; fundamental; controlled; antiresonance.

Inches	mm
.002	0.05
.005	0.13
.008	0.20
.010	0.25
.015	0.38
.020	0.51
.025	0.64
.030	0.76
.035	0.89
.040	1.02
.050	1.27
.075	1.91
.141	3.58
.238	6.05
.317	8.05
.352	8.94
.486	12.34
.725	18.42
.757	19.23
.775	19.69



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking shall be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR132/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: HC-6/U (MIL-H-10056/2).

Dimensions, marking, and configuration: See figure 1.

Frequency range: 0.8 to 20 MHz, inclusive.

Frequency tolerance: ±20 ppm.

Antiresonance, load capacitance: 30.0 ±0.5 pF.

Frequency and equivalent resistance:

Frequency range, inclusive MHz			Maximum resistance ohms
0.8	to	0.85	630
0.85+	to	0.9	600
0.9+	to	1	580
1+	to	1.12	540
1.12+	to	1.25	490
1.25+	to	1.37	450
1.37+	to	1.5	410
1.5+	to	1.62	380
1.62+	to	1.75	330
1.75+	to	1.87	300
1.87+	to	2.0	290
2.0+	to	2.12	270
2.12+	to	2.25	250
2.25+	to	2.6	200
2.6+	to	3	150
3+	to	3.4	110
3.4+	to	3.75	90
3.75+	to	4	75
4+	to	5	60
5+	to	7	35
7+	to	10	24
10+	to	15	22
15+	to	20	20

Mode of oscillation: Fundamental.

Operatable temperature range: -55°C to +80°C, inclusive.

Operating temperature range: +80°C to +90°C, inclusive.

Drive level: 0.8 to 10.0 MHz = 5.0 ±1.0 mW.

10.0+ to 20.0 MHz = 2.5 ±0.5 mW.

Capacitance, shunt: 7.0 pF maximum.

Shock (specified pulse): 1/

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: 0.8 to 2.0 MHz ±15 percent.

2.0+ to 20.0 MHz ±10 percent.

1/ The violation of $P = I^2R$ is intentional; at the higher resistance of the crystal unit.

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Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: 0.8 to 2.0 MHz ±15 percent.

2.0+ to 20.0 MHz ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Quality assurance provisions:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A tests of MIL-C-3098 shall be applicable.

NOTES:

1. The socket designed to mate with crystal holder HC-6/U is part no. M12883/04-01 (see MIL-S-12883/4). For Navy use, other sockets listed in MIL-STD-242 may be used.
2. NATO preferred type NXT 12/B.
3. This crystal unit is identical to the CR36A/U, except for a load capacitance rating of 30 ±0.5 pF. This unit is not interchangeable with the CR36A/U.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, SM
Navy - SH
Air Force - 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0672-08)

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